

Guitar Effects Chain



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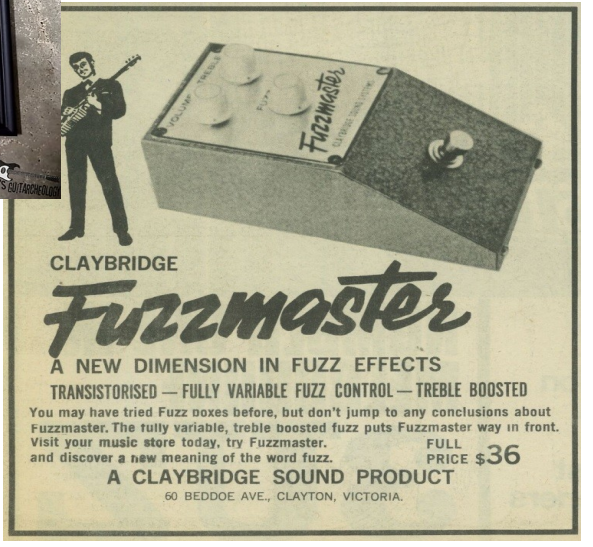
Some old ads











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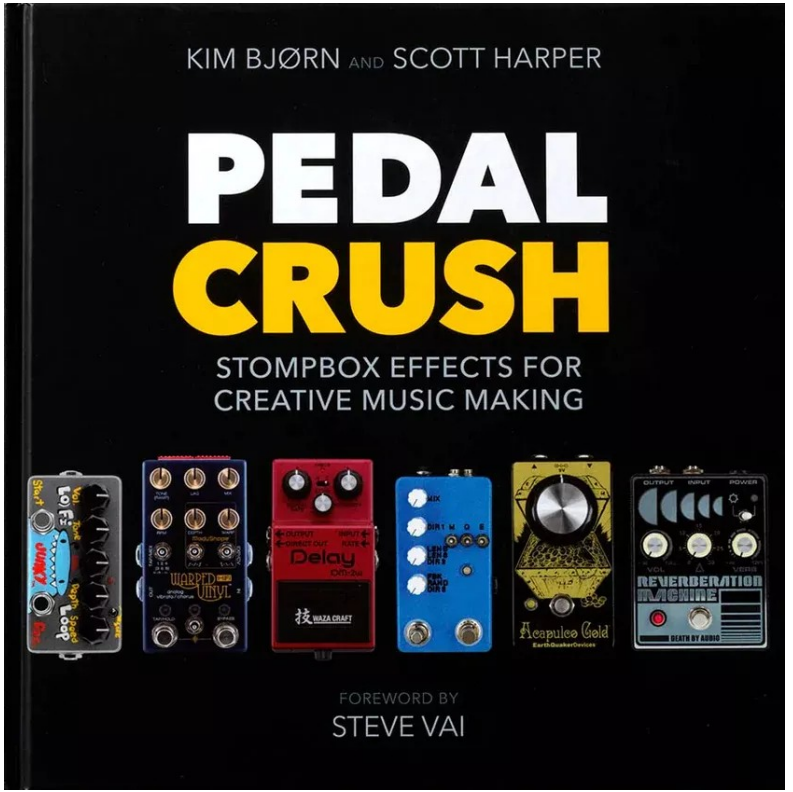


References

Book about pedals, pedals and people building pedals :

Pedal Crush

A lot of good information in this book



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Plugins used

- **mod-pitchshifter / mod-gxpitchshifter** : pitch shifting plugins. The gx version is a modified version of mod-pitchshifter useable with Guitarix
- **spectra.lv2** : A spectrum analyser from x42 as a LV2 plugin
- **sisco.lv2** : an oscilloscope from x42 as a LV2 plugin
- **lsp-plugins-lv2** : some LV2 plugins. A nice spectrum analyzer is available in the package
- **guitarix** : a lamp amplifier simulator. A complete Guitar amplifier with nice effects
- **lv2-calf-plugins-gui** : some nice plugins. We will use the compressor
- **raysession** : a jack session manager
- **audacity** : a powerful sound recorder
- **Carla** : a plugin rack. We will use carla to manage the oscilloscope and the spectrum analyzer
- **Zam** : a collection of audio plugins for high quality processing

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Guitar Effects Chain



Pre-Amp

Pitch

Comp.

Dist.

Wah

Mod.

Delay

Eq.

Gate

Volume

Reverb





Guitar Effects Chain

Defining a good guitar effects chains is hard :
- there is no general rule
- you can put the noise gate at the beginning or after the distortion
- you can put the wah pedal at the beginnig of after the distortion
- you can put the equalizer everywhere you can (but it has a good job to do at the beginning)
- the distortion you put before the head of the amplifier is just used as an amplifier, the drive level has nearly no effect

Follow a general schema, but try to move elements, it can be best for you

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MIDI Pedal Board



Line6 FBV
Express MKII



MIDI Mouse
Tech 21



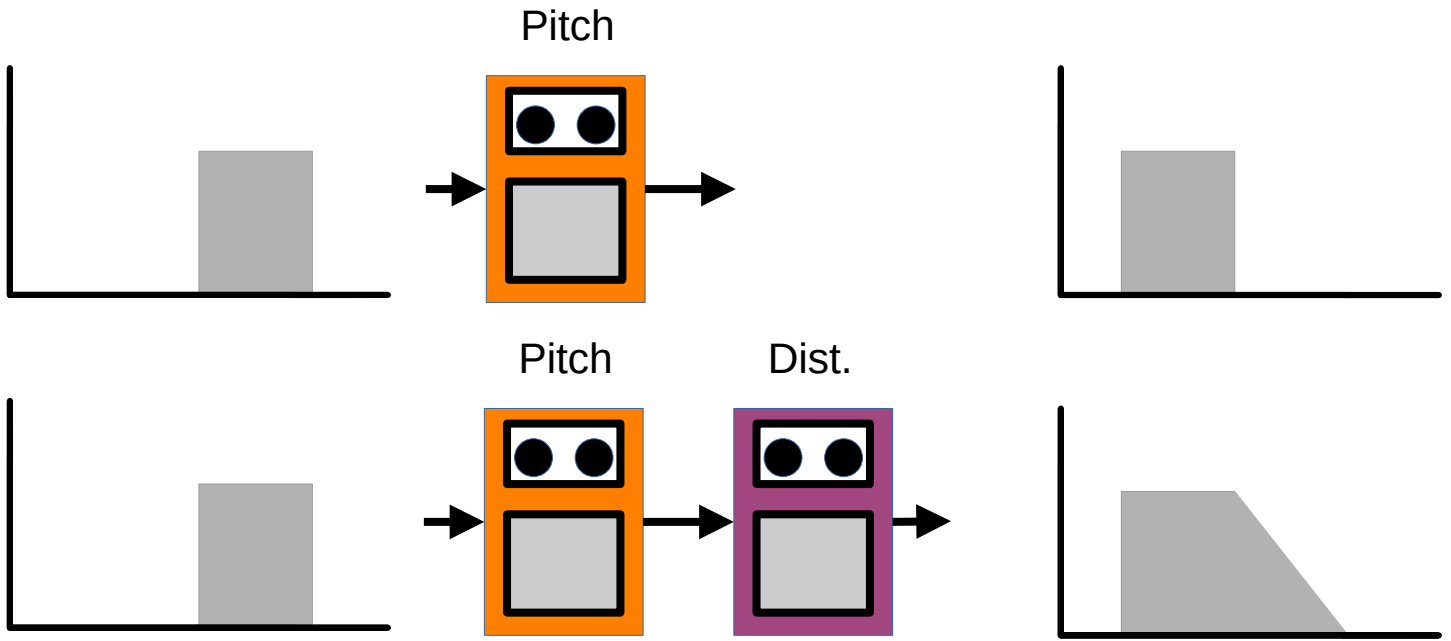
Behringer
Fcb1010

Some interesting MIDI pedal board used to control Guitarix or other applications

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Pitch Shifter Effect 1/2



Adding a small level of distortion after a pitch shifter can add a some missing pieces of frequencies when downtuning a guitar

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Pitch Shifter Effect 2/2



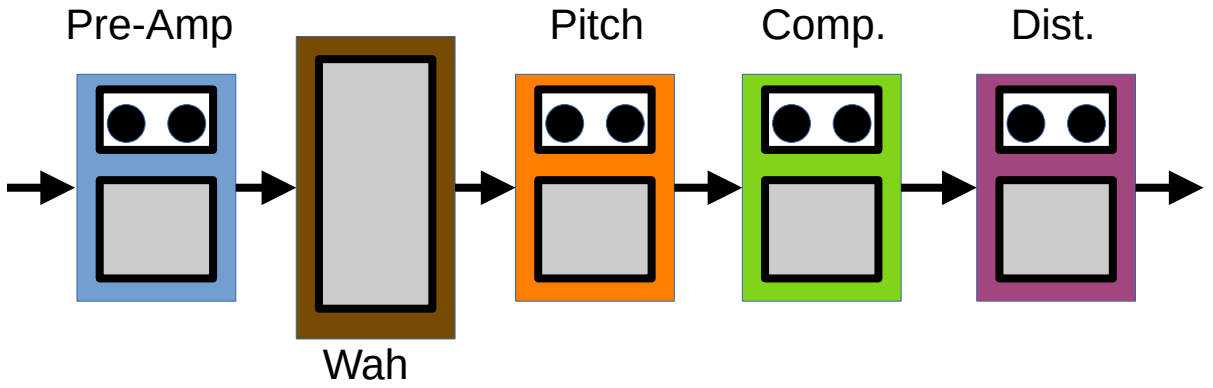


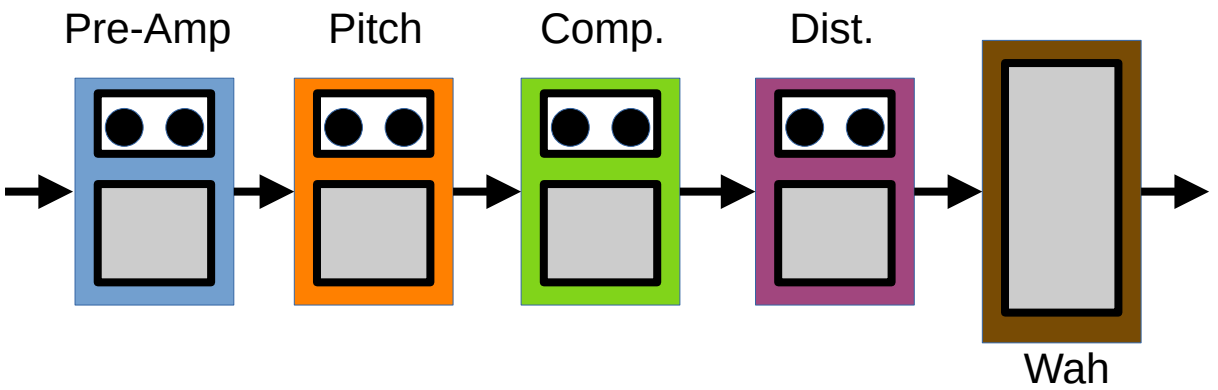
GxCapo : a LV2 plugin used to tune up the guitar
GxDrop : a LV2 plugin used to tune down the guitar
Overdrive : an LV2 used to add a little distortion to help fill the missing bits in the spectrum

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Wah Pedal Position





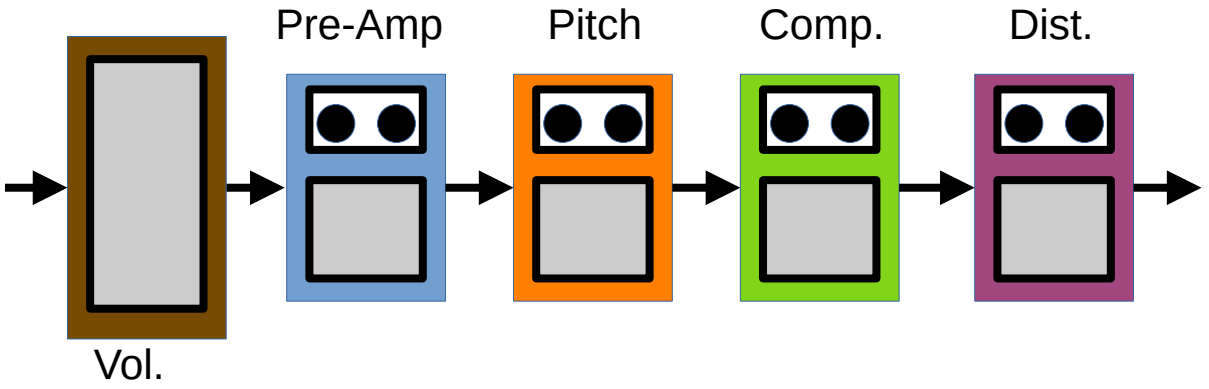
In the pre-amp we have a quasi-sinusoïd from the guitar.
The wah pedal which is a filter will just filter a quasi-sinusoïd and will behave more or less as a volume pedal.

After the distortion pedal, the signal has a richer specter and when processed by the wah, the wah effect will be more pronounced.

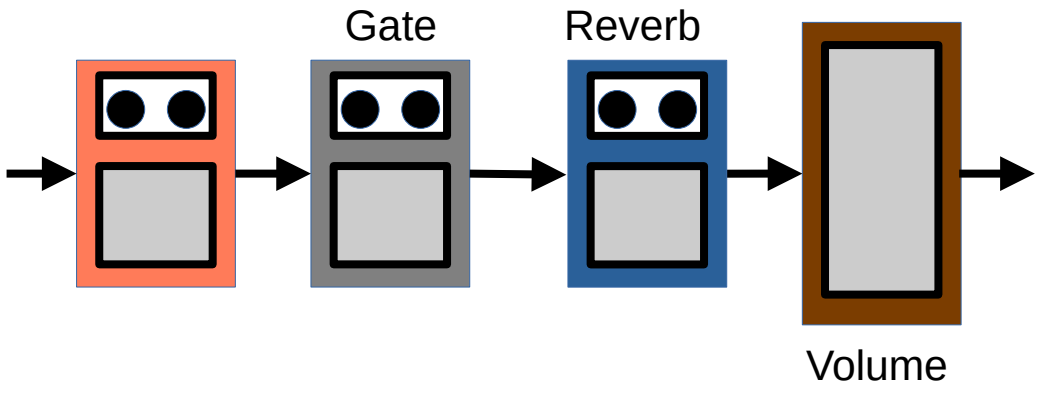
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Volume Pedal Position



Placed at the beginning of the chain, the volume can have an impact on the sound. Compressor and distortion are tuned for a given level of signal.

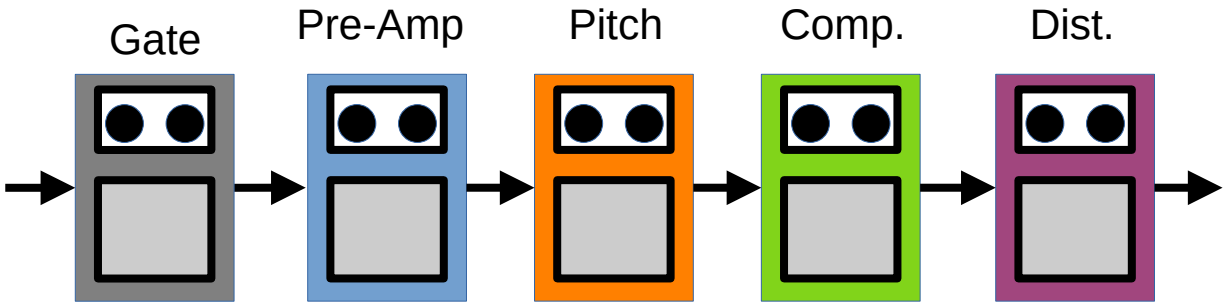


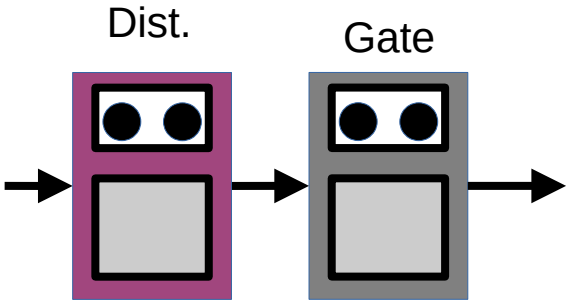
At the end of the chain, the identity of the sound will be preserved if we reduce the level of signal via the volume pedal.

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Noise Gate Position





Placed at the beginning of the chain, the volume can have an impact on the sound. Compressor and distortion are tuned for a given level of signal.

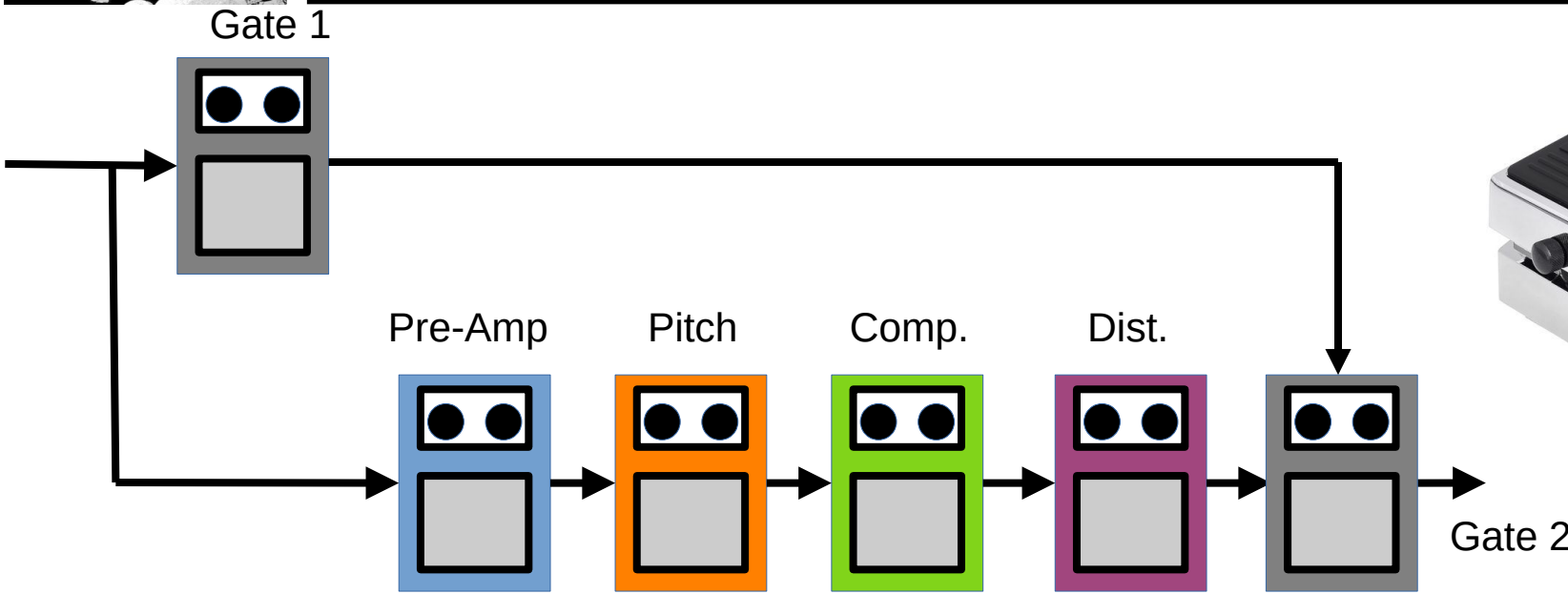
At the end of the chain, the identity of the sound will be preserved if we reduce the level of signal via the volume pedal.

… Or both. At the beginning (for guitar mic noise) and at the end (for pedal noise).

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Sidechained Noise Gate 1/2



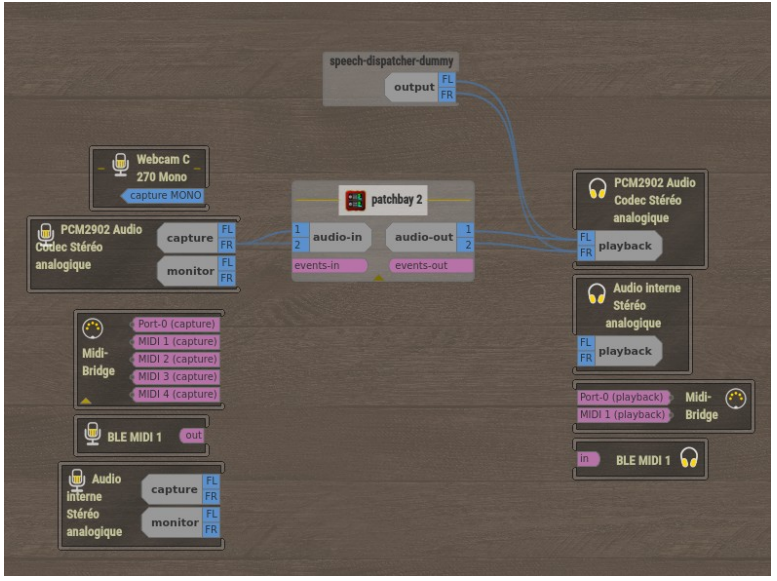


The sidechained noise gate will get the enveloppe from the input signal (Gate1) and will use this enveloppe to cut off the processed signal at the end of the effect chain. So, it will remove the noise produced by all the pedal using information from the beginning of the effect chain (check the **ISP Technologies Decimator X G-String Noise Red**).

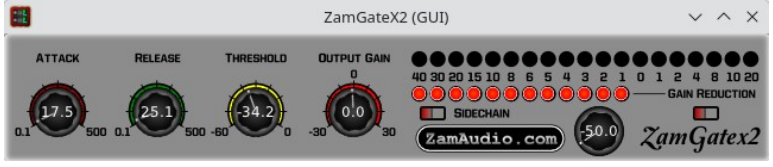
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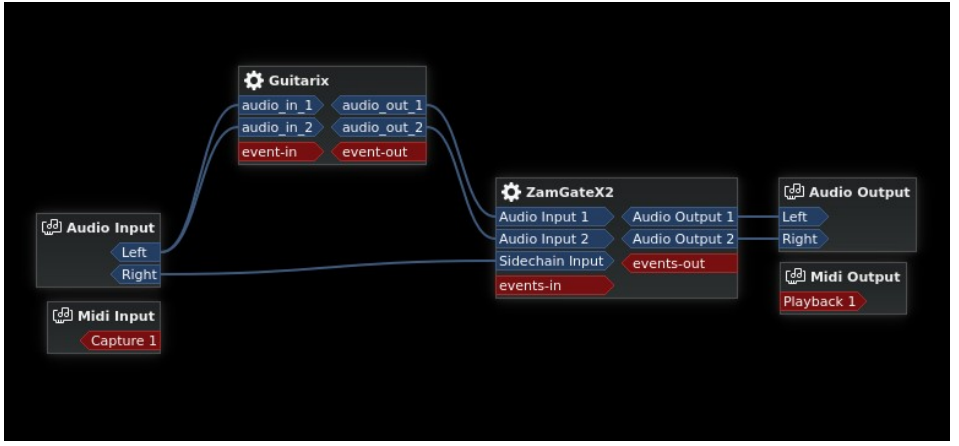


Sidechained Noise Gate 2/2



The connexions in Ray Session





The connexions in carla-patchbay :
- Guitarix launched as application
- ZamGatex2 loaded as a plugin
- carla-patchbay loaded in Ray Session

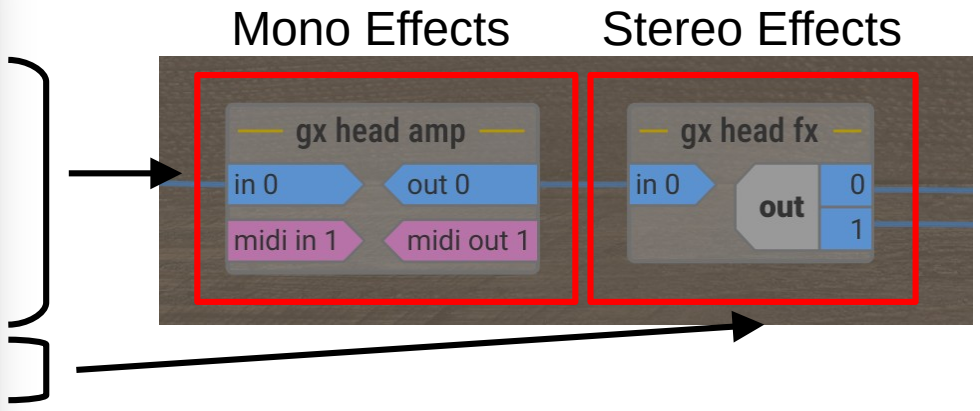
The ISP Decimator (289 € on Thomann) do the same thing as ZamGate. Here is an example from Tosin Abasi pedalboard : Abasi / YouTube

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Guitarix Rack Structure





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Guitarix Default Racks

Level of the threshold compressor at the top of the rack

Clipping level at the bottom of the rack



Level of the noise gate at the top of the rack

Gain of the preamp at the bottom of the mono rack

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Guitarix Default Racks

Gain at amp input

Gain at amp output

Lamp configuration selection

Allow freq below 31 Hz



Mix level

Gain of the lamp stage

Bass level

High frequency boost

Amp internal reverb level

Hiss remover

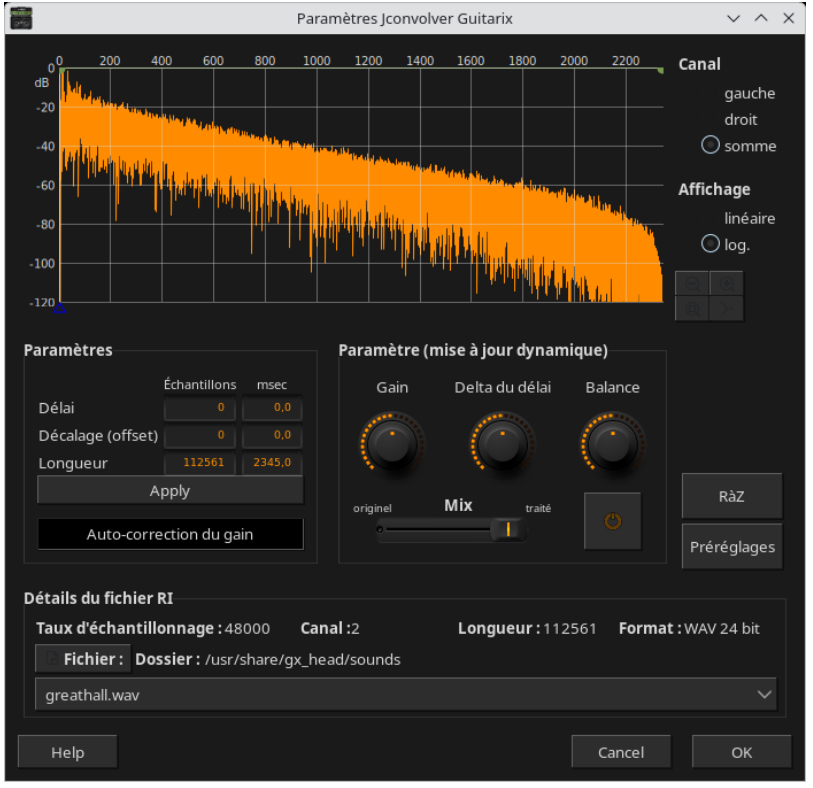
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Guitarix Default Racks

Convoler : to copy the style of a given band
Amp Impulse : to copy an amplifier style





Convolver Window Setting

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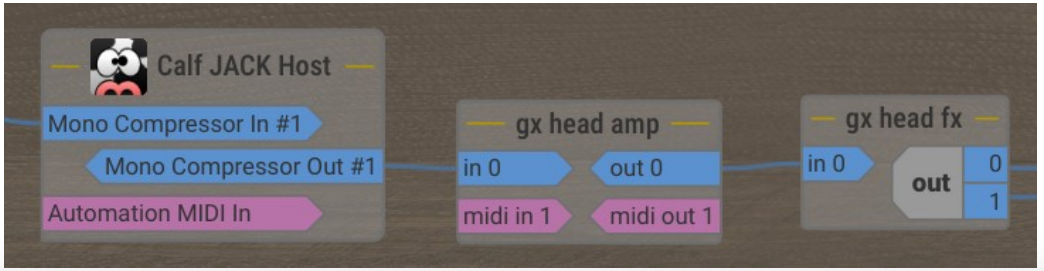


Compressor Experiment

We use the compressor from Calf plugins. This one has a nice GUI which eases the tuning of the compressor.

We then use audacity to generate a special signal using Generate -> Silence + Generate -> Tone to show the compressor behavior.

This signal is sent to calf / compressor and audacity records the audio of calf / compressor





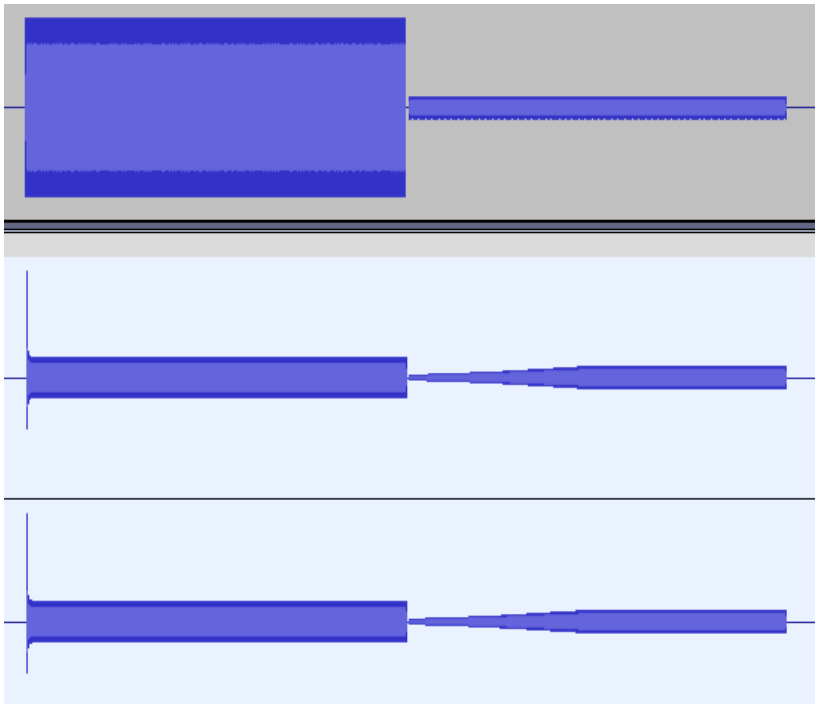


Compressor Experiment

The first part of the signal is above the threshold and the second part is below.
I set a huge release and a huge attack level.

With the attack set to a non nul value, the output signal start with the original level and then reach the level defined by the compressor at the speed defined by the attack level/

With the release set to a non nul value, at the amplitude change, the compressor keeps its gain and apply the gain to the low amplitude and slowly reach the unity gain and the speed defined by the release level.



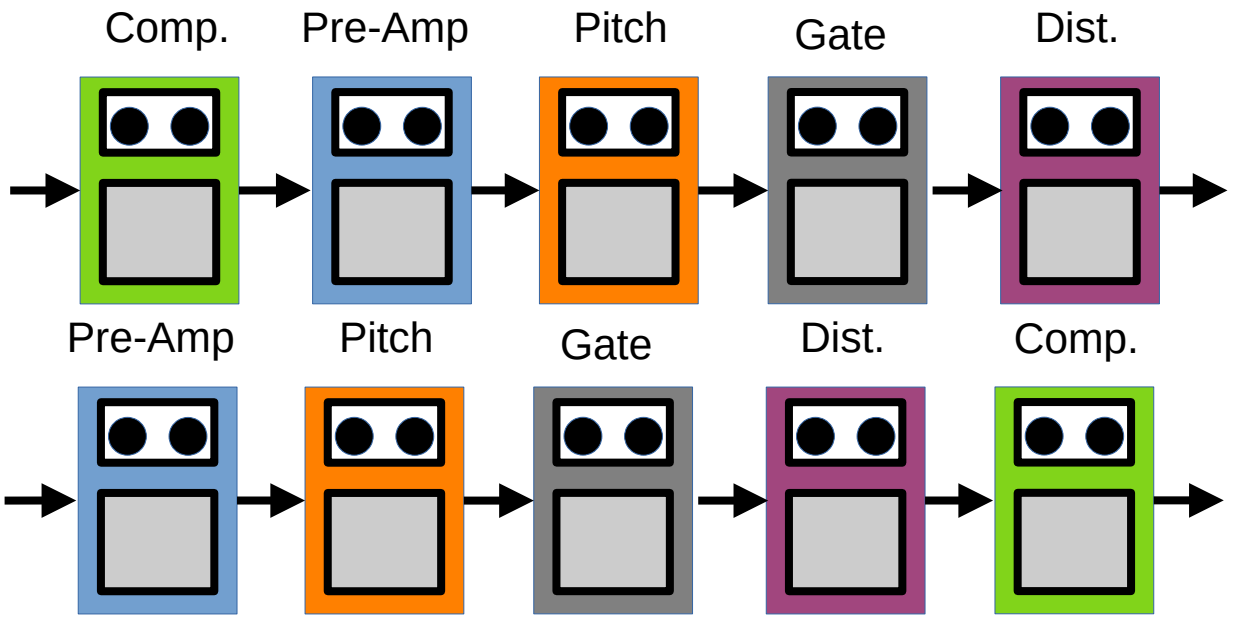
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Compressor

How to use the compressor :
- you can use a compressor at the beginning of the signal chain. In this position, it is mostly used for sustain.
- you can use it after the « last » effect in the chain (like after the distortion). In this position, it will be mostly efficient for adjusting the sound for tapping / sweeping for example.

Example of Guitarix / Compressor settings for tapping / sweeping :
Ratio : 2.5
Threshold : -18
Knee : 6
Attack : 0.01
Release:0.25



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Compressor : Tuning

How to tune the compressor :
- Play some tapping or legato notes and set the threshold so that the notes are above the threshold in the Calf compress graph
- Play some notes using the pick and set the compressor level to reduce the level difference between a legato or tapping note and the note played with the pick.



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Compressor : LSP

LSP Compressors are good.
There are many available :

- Compressor LeftRight
- Compressor MidSide
- Compressor Mono
- Compressor Stereo
- Sidechain Compressor LeftRight
- Sidechain Compressor MidSide
- Sidechain Compressor Mono
- Sidechain Compressor Stereo



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Distortion

Overdrive
Overdrive simulates an amp pushed to its limits, and early overdrives (like the Tube Screamer) were designed to push the preamp stage of an already hot tube amp into natural saturation.

But an overdrive itself produces slight distortion - it's simply a smooth, asymmetrical saturation that clips the signal gradually and organically, while retaining a lot of dynamics. Play softly → almost clean sound. Play hard → natural saturation.
This responsiveness to touch is what characterizes it.

Distortion
A distortion pedal clips the signal symmetrically and aggressively, often to the point of total clipping (square). The result is denser, more consistent, less responsive to touch - the sound is saturated regardless of how hard you play.

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Distortion

	Overdrive	Distortion
Clipping Type:	Asymmetrical, smooth.	Symmetrical, aggressive.
Dynamic:	Very touch-sensitive.	Not very responsive.
Harmonics:	Even (2nd, 4th) → warm.	Odd (3rd, 5th) → aggressive.
Amp Interaction:	Strong	Weak

The real rule

It's not so much the final stage that matters, but rather the frequency coloration introduced at each stage that survives. Cascading distortion erases dynamics but preserves tonal balance.

This is why guitarists often layer an overdrive before a distortion pedal before a saturated amp - each stage colors the spectrum differently, even if the dynamics are compressed from the very first stage.

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Distortion

Why the final stage dominates :

Distortion is a destructive process - once the signal is clipped, the original dynamic information is lost. No matter what you add upstream afterward, you're working with an already depleted signal. This is the principle of cascading non-linearities: if your amp saturates heavily, it will re-distort everything that comes in, whether it's already distorted or not.
The final saturation stage imposes its characteristic tone.

But that's not entirely accurate.
Several things happen beforehand that survive the final distortion:

- **Equalization**: the Tube Screamer cuts the bass frequencies before distortion - this characteristic mid-boost remains audible even after a saturated amp, because distortion amplifies differently depending on the frequencies present.
- **Input level**: the more input you send to the amp, the more you push it into its own saturation - this is the whole point of a boost or overdrive in front of a tube amp.
- **Residual dynamics**: a slightly overdriven signal still retains some dynamic range, which the amp will process differently than a clean signal.

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Distortion

Use scopes to see the shape of the signal as you add effects (compressor, distortion, an overdrive, etc …)

Use spectrum analyzer to check the shape of the spectrum :

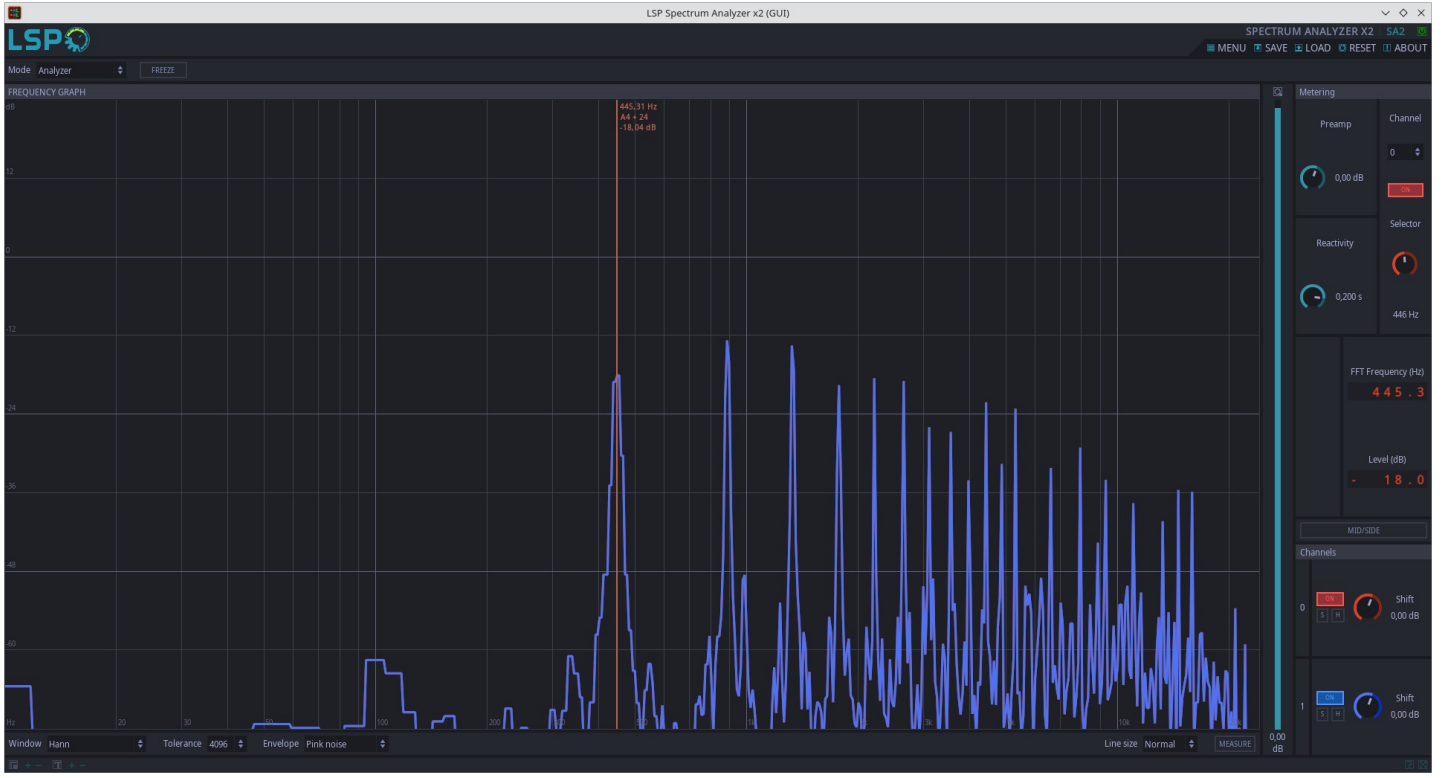
- Clipping
- Too much distortion may produce high frequencies
- 50Hz from power supply may produce parasitic frequencies like 200 Hz peak

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Distortion

Here is a A (440 Hz) played on a guitar with a lot of distortion.
As you can see the first harmonic has a level above the fundamental.
The content of the spectrum as a lot of high frequencies and you may need to reduce a little bit the distortion to keep some limits on higher frequencies.



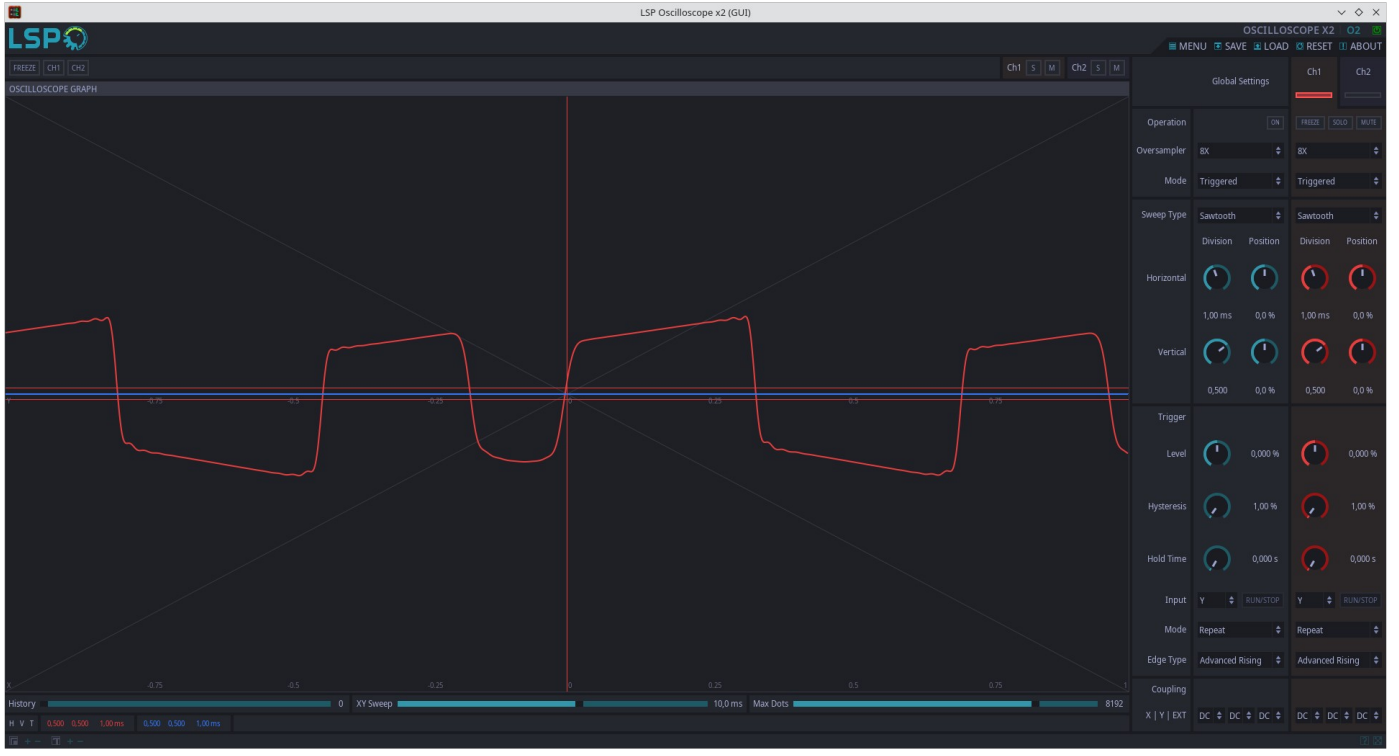
LSP spectrum analyzer

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Distortion

Use a scope to check the shape of the sound and see how parameters can change the sound

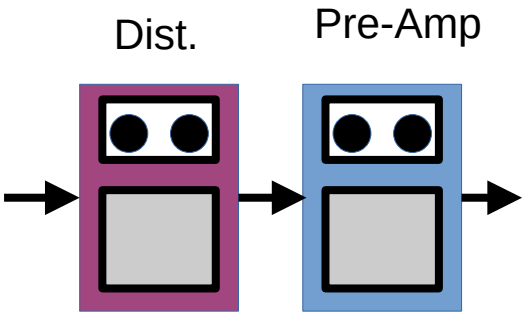


LSP Oscilloscope

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Some Distortions tuning



When using a distortion before the pre-amp or the amplifier head, the first drive / tone distortion tuning has nearly no effects. Use a little drive in the first distortion.
For the second one (the pre-amp or the amplifier head), use the level of drive you want, but after that, to limit the aliasing effects due to the spectre extension produced by the distortion, use a low pass filter to limit the frequencies after the Nyquist frequency.

Aliasing can produce some unwanted noises like ike crackling, harshness and metallic noises.

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Example of Guitarix settings

Overdrive : 5.1 / 100 %
Compressor : 4.5 / -51.1 / 2 / 0 / 0
Rat : 0.86 / -4.8 / 0.75
Amp : See after
Wah : ColorSound Wah / 0 / 52
Echo : 150 / 15.0
Volume : 6.3
Freeverb : 0.78 / 0.50 / 40
Cabinet : 4x12 / -0.5 / 0.6 / 3.8



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Example of Guitarix settings



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AIDA-X



AIDA-X are neural amplifier models developed for the MOD Device. These models a lightweight enough to be used in real time.
You can load an AIDAI-X model via the RTNeural Guitarix plugin.
The LV2-AIDA-X plugiin is not yet compatible with Guitarix but can be loaded into Carla-Rack.
You can find some good AIDA-X models on :
https://www.tone3000.com/

On this web site you can also find IR of cabinets.

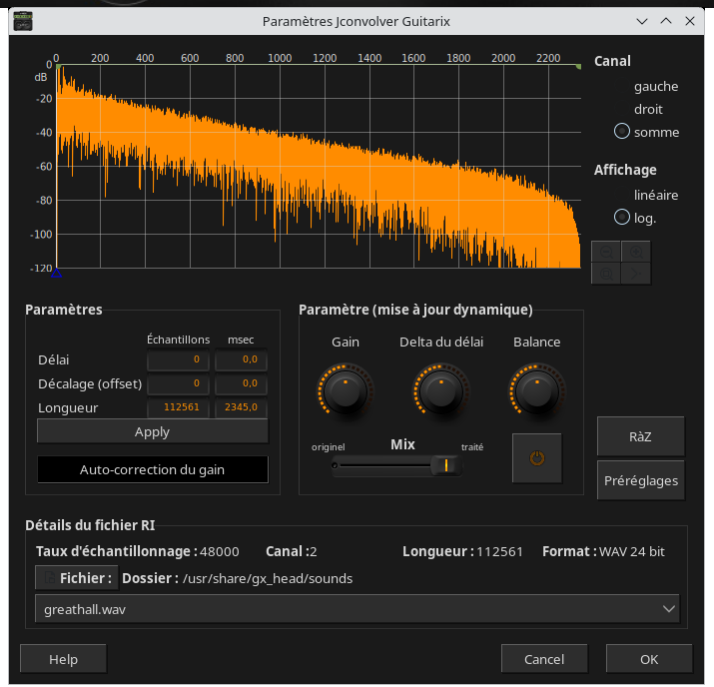
The NAM model is quite big and when I tried it, the load was too high and produced X-runs.

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Convolver





The convolver allows you to applied an Impulse Response (IR) on an input signal. The tool allows to ‘copy‘ a given sound.

With guitarix, you can find a lot of preinstalled IR for bands and for amplifier.

Another usage of IR is to reproduce a reverberation. You can find a lot of IR for reverberation on the internet.

The last usage is to mimic a cabinet (and ensemble of speakers). Same as for reverberation, you can find a lot of IR for cabinets on the internet.

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NAM

Neural Amp Modeler is an open-source project that uses deep learning to create models of guitar amplifiers and pedals with state-of-the-art accuracy.

There are 2 NAM version :

- A1 : the old standard

- A2 : a new version with a new parameter controlling the level of accuracy and so, the CPU load required for evaluation

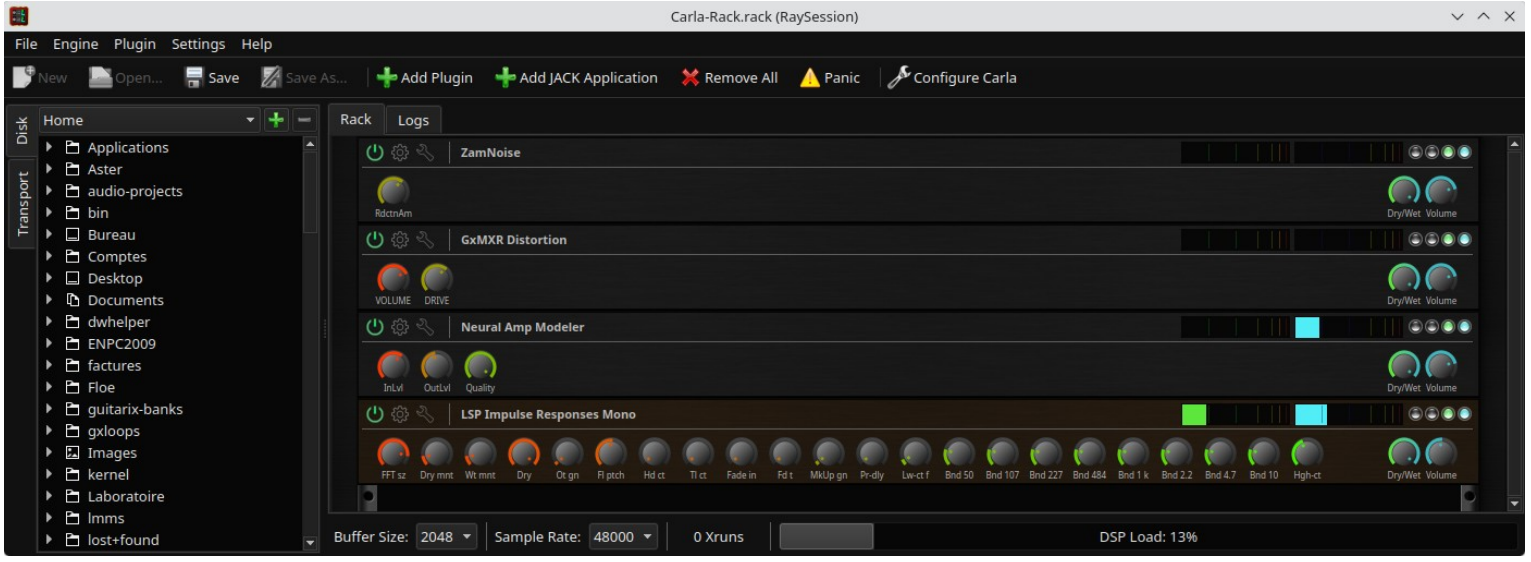
Both models are available now on Fedora :

- lv2-neural-amp-modeler
- lv2-neuralrack
- mod-neural-amp
- a2core
- ratatouille

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NAM A2



This preset uses :

- **ZamNoise** : as a noise gate
- **GxMXR Distortion** : as an overdrive for the NAM modl
- **Neural Amp Modeler** : as a NAM A2 loader (model downloaded from https://www.tone3000.com)
- **LSP Impulse Response Mono** : as an IR loader for the Cabinet simulation

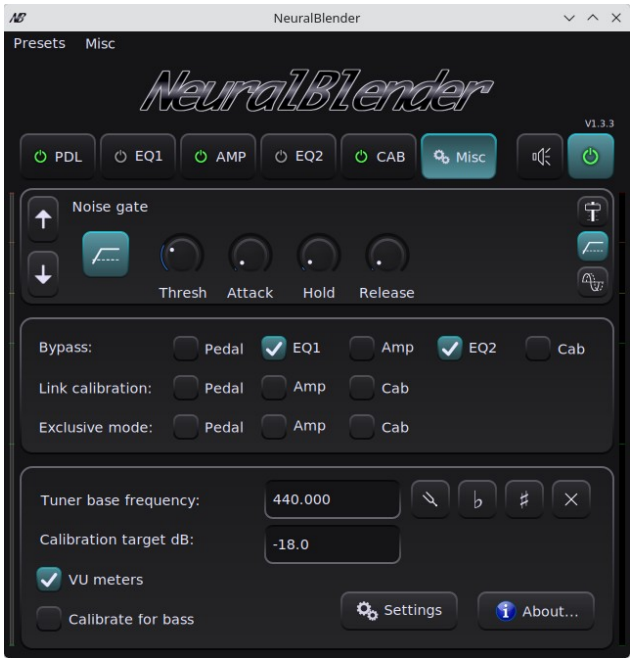
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NeuralBlender



A tool which allow to mix / blend several NAM A2 / IR models for amplifier / cabinets and effects.



https://sourceforge.net/projects/neuralblender/

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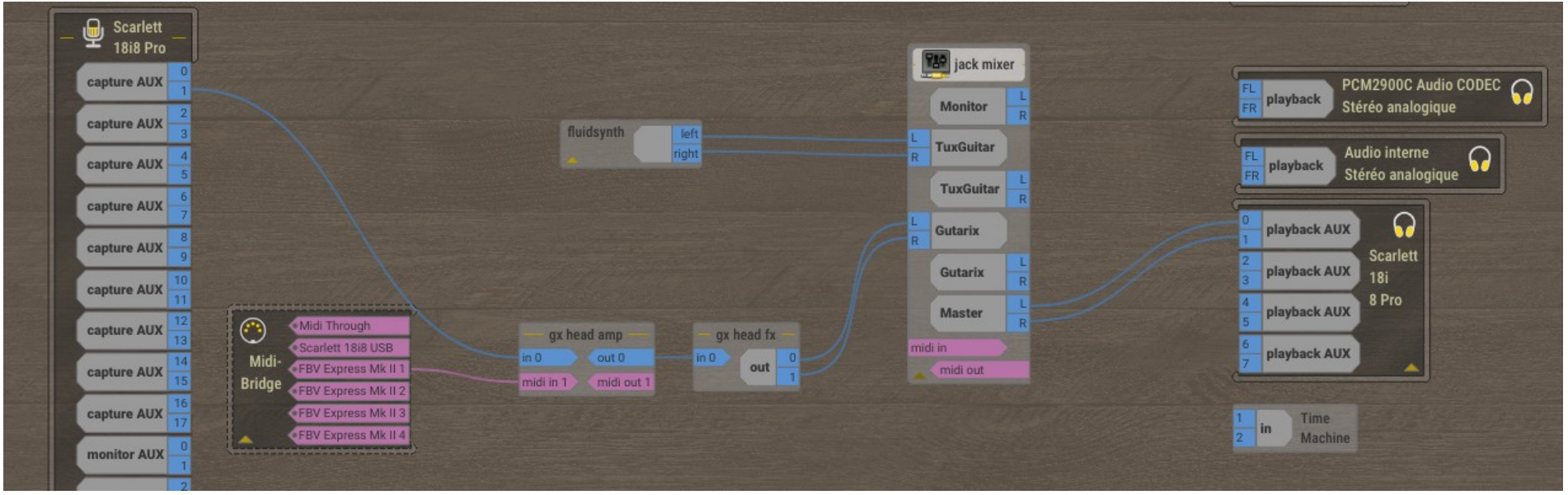
Rehearse with TuxGuitar

Applications used :
- **guitarix** : guitar amplifier simulator
- **tuxguitar** : to play GuitarPro scores
- **timemachine** : to record some audio
- **jack_mixer** : to fine tune the audio level between tuxguitar and guitarix
- **non-mixer-xt** : (another option to replace jack_mixer)
- **ray session** : to start all these applications and reconnect them

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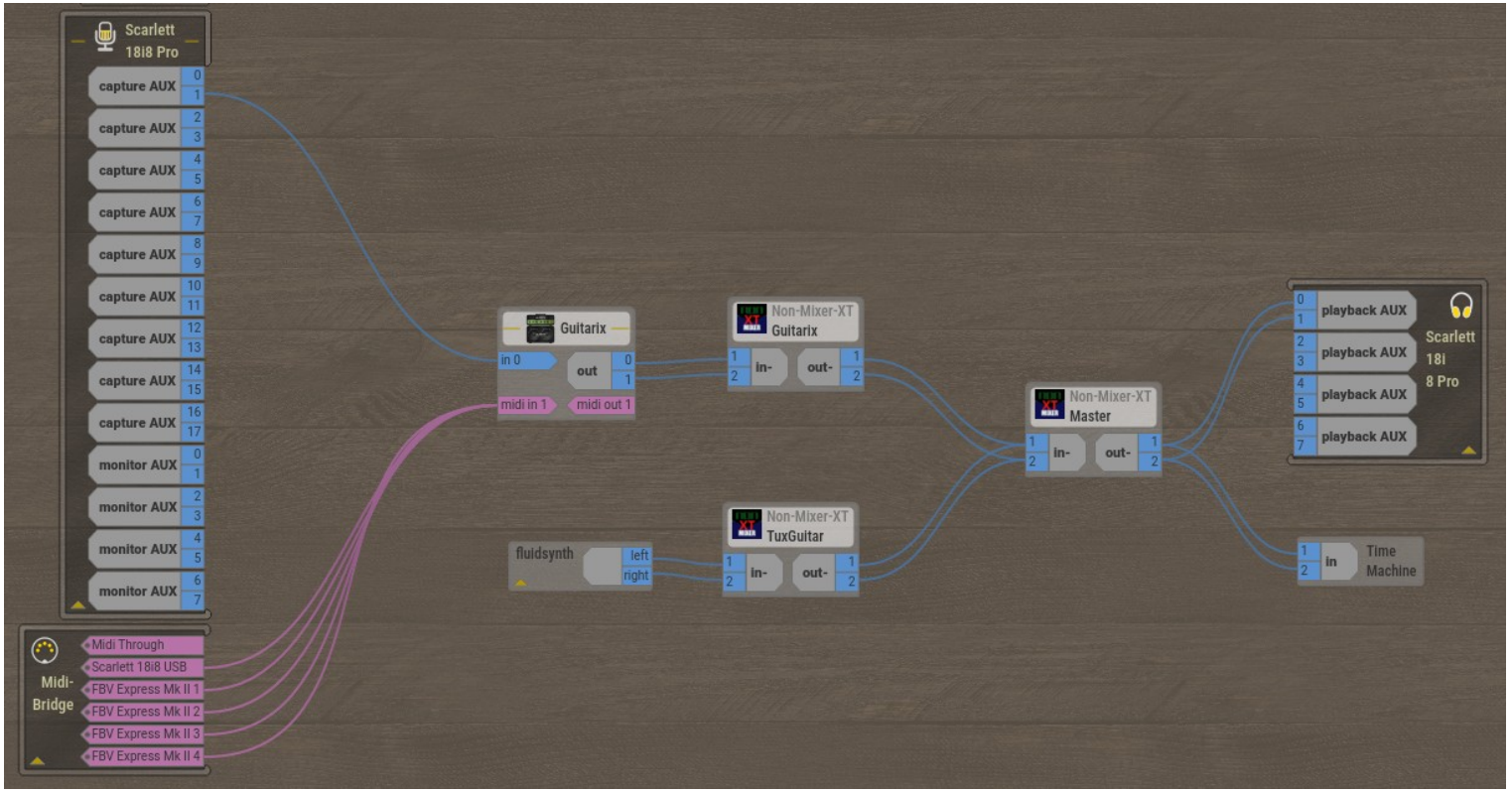
Rehearse with TuxGuitar



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Rehearse with TuxGuitar - 2

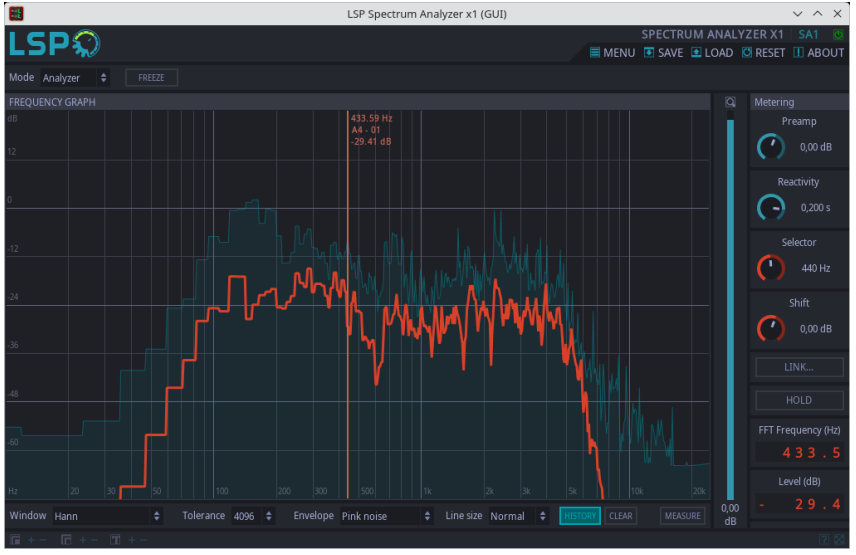


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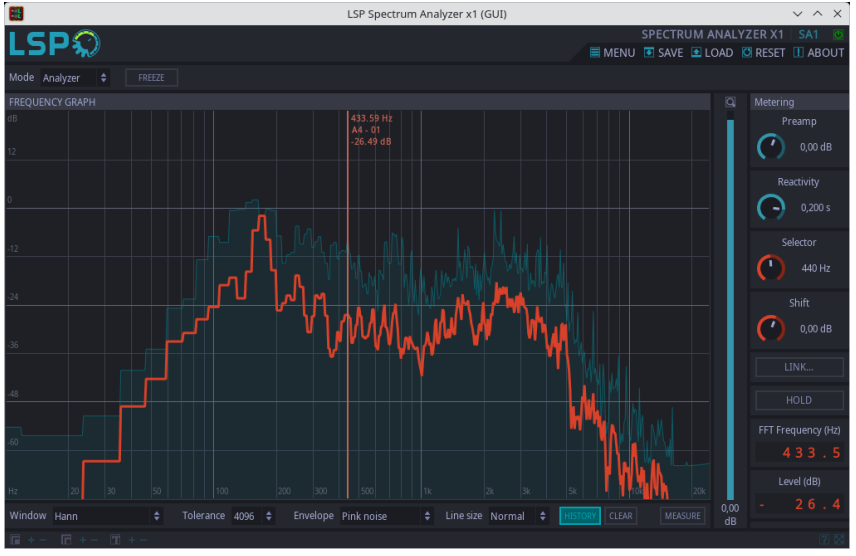


Spectral Comparison of Sounds

Sound with standard Guitarix distortion settings



Sound with AIDA-X model and high gain



The spectral pic at 2500 Hz is due to the IR of the cabinet. Pic not present with another cabinet

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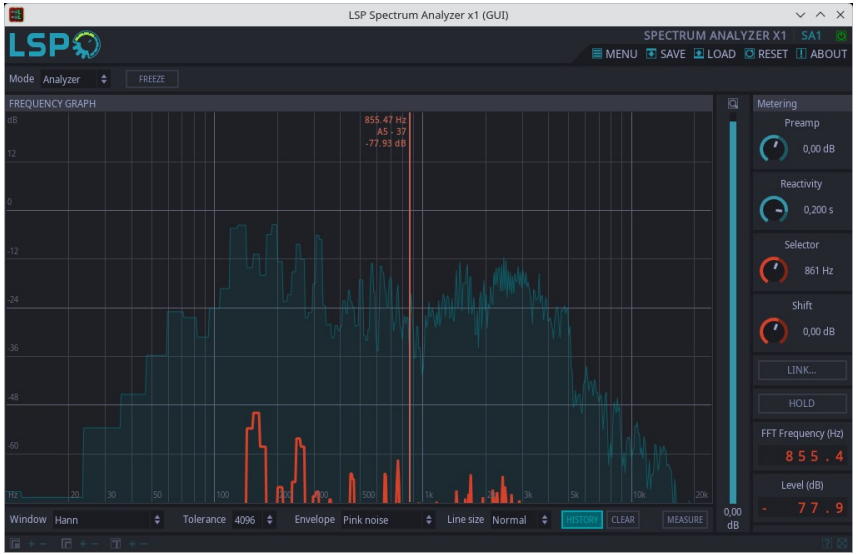
Spectral Comparison of Sounds

Spectrum with a humbucker coil



A small peak at 50 Hz and a small peak at 2500 Hz

Spectrum with a single coil



A big peak at 50 Hz with some harmonics and a bigger peak at 2500 Hz

The n x 50 Hz harmonices are due to power supply + distortion

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Additional effects for Solo

We can add to the current settings :
- a **volume** pedal adding +3 to +6 dB
- an **echo** with a delay between 300-450 ms or 120-250 ms for a subtil echo and add a feedback between 20-35 %. The wet/dry mix is around 15-25 %. The echo is placed before the reverb. ChatGPT suggest to add a high pass filter with a cutting frequency between 150-200 Hz.

For a compressor for tapping / standard play :
Threshold : around -20 dB (adjust to correspond to the signal level – lower this value until the tapping trigger compression).
Ratio : 3:1 to 4:1.
Attack : 10 to 20 ms → let pass the mediator attack to have a percussive sound.
Release : 80 to 150 ms → fast enough so that compression breathe between note.
Makeup gain : tune so that tapping and mediator play are at the same level.
Knee : soft knee if available → the compression acts softly and avoid pumping effect.

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